

Land Use / Land Cover (LULC) Classification

Experiment 2: Spectral Indices and Temporal Feature Engineering

Random Forest vs. XGBoost — Sentinel-2 Spectral Indices & Temporal Features

Project Report

1. Objective

The objective of this experiment was to evaluate the effectiveness of feature engineering for LULC classification by combining raw Sentinel-2 spectral bands with spectral indices and temporal difference features. This experiment aimed to determine whether additional information derived from vegetation, moisture, built-up, and soil-related indices could provide a more discriminative feature representation.

2. Dataset and Feature Configuration

Parameter	Configuration
Experiment ID	EXP-02
Experiment Type	Feature Engineering
Input Data	Sentinel-2 multispectral data
Number of Seasons	3
Seasonal Features	S1, S2, S3
Raw Spectral Bands	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12
Raw Band Features	30
Spectral Indices	BSI, EVI, NDBI, NDMI, NDVI, NDWI, SAVI
Spectral Index Features	21
Temporal Difference Features	NDMI_diff_12, NDMI_diff_23, NDVI_diff_12, NDVI_diff_23
Temporal Difference Feature Count	4
Total Input Features	55
Number of Classes	7
Target Samples per Class	~3,000
Total Samples	20,963
Test Samples	4,193 (~20%)
Class Labels	0-6
Original Class Labels	10, 20, 30, 40, 50, 60, 80

Feature Composition

The 30 raw spectral features were obtained from 10 Sentinel-2 bands across three seasons. Seven spectral indices were calculated for each season, resulting in 21 additional features. Temporal changes were represented using NDVI and NDMI differences between consecutive seasons, contributing four additional features.

Feature Group	Number of Features
Raw spectral bands	30

Feature Group	Number of Features
Spectral indices	21
Temporal difference features	4
Total	55

3. Class Distribution

The original class labels were remapped as follows:

Original Label	Remapped Label	Samples
10	0	3,000
20	1	2,972
30	2	3,000
40	3	3,000
50	4	3,000
60	5	2,991
80	6	3,000

The dataset was therefore approximately balanced, with each class containing close to 3,000 samples.

4. Models Evaluated

Two ensemble-based machine learning classifiers were evaluated using the same 55-feature dataset:

- Random Forest Classifier
- XGBoost Classifier

This enabled comparison of the two models under the same feature-engineering configuration.

5. Overall Results

Model	Accuracy	Macro Precision	Macro Recall	Macro F1	Weighted F1
Random Forest	87.00%	0.87	0.87	0.87	0.87
XGBoost	88.00%	0.88	0.88	0.88	0.88

Best Performing Model

XGBoost achieved the best performance:

- Accuracy: 88.00%
- Macro Precision: 0.88
- Macro Recall: 0.88
- Macro F1-score: 0.88
- Weighted F1-score: 0.88

XGBoost outperformed Random Forest by 1 percentage point in accuracy and 0.01 in Macro F1-score.

6. Random Forest Class-wise Performance

Class	Precision	Recall	F1-score	Support
0	0.82	0.82	0.82	600
1	0.87	0.88	0.88	586
2	0.78	0.82	0.80	574
3	0.89	0.92	0.90	583
4	0.84	0.80	0.82	629
5	0.91	0.87	0.89	631
6	0.98	1.00	0.99	590
Macro Avg	0.87	0.87	0.87	4,193
Weighted Avg	0.87	0.87	0.87	4,193

7. XGBoost Class-wise Performance

Class	Precision	Recall	F1-score	Support
0	0.81	0.83	0.82	582
1	0.90	0.87	0.89	615
2	0.80	0.84	0.82	574
3	0.91	0.93	0.92	583
4	0.85	0.83	0.84	617
5	0.92	0.88	0.90	627
6	0.99	0.99	0.99	595
Macro Avg	0.88	0.88	0.88	4,193
Weighted Avg	0.88	0.88	0.88	4,193

8. Key Observations

- The combination of raw spectral bands, spectral indices, and temporal features provided a strong representation for LULC classification.
- XGBoost performed better than Random Forest, achieving 88.00% accuracy compared with 87.00% for Random Forest.
- XGBoost achieved a Macro F1-score of 0.88, indicating strong and relatively balanced performance across the seven classes.
- Class 6 was the best-performing class for both models, achieving an F1-score of 0.99.
- For XGBoost, Classes 3 and 5 also showed strong performance, with F1-scores of 0.92 and 0.90, respectively.
- Class 2 remained comparatively challenging, although XGBoost achieved an F1-score of 0.82.
- The similarity between Macro F1 and Weighted F1 indicates that the approximately balanced class distribution did not substantially distort the overall evaluation.

9. Conclusion

Experiment 2 investigated the effect of feature engineering by combining raw Sentinel-2 spectral bands with spectral indices and temporal difference features. The resulting feature space consisted of 55 features derived from three seasonal observations.

Seven spectral indices — BSI, EVI, NDBI, NDMI, NDVI, NDWI, and SAVI — were incorporated along with temporal changes in NDVI and NDMI. Random Forest achieved an accuracy of 87.00%, while XGBoost achieved the best performance with an accuracy of 88.00% and a Macro F1-score of 0.88.

These results demonstrate that the engineered feature representation provides substantial information for distinguishing different LULC classes, with XGBoost showing slightly better performance than Random Forest under this configuration.